

| Power Budget for HMI - Jacob Alger                                                                                                                                |                                                                                                        |                     |                    |   |                                                |                  |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------|--------------------|---|------------------------------------------------|------------------|------|
| Team Number:                                                                                                                                                      | Team 202                                                                                               |                     |                    |   |                                                |                  |      |
| Project Name:                                                                                                                                                     | Cropscout                                                                                              |                     |                    |   |                                                |                  |      |
| Team Member Names:                                                                                                                                                | Jacob Alger, Caleb Yuen, Aaron Kiem, Isaiah Johnston, Cristopher Gutierrez, Asadbek Ruziev, Tyler Dean |                     |                    |   |                                                |                  |      |
| Version:                                                                                                                                                          | 2                                                                                                      |                     |                    |   |                                                |                  |      |
| A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements |                                                                                                        |                     |                    |   |                                                |                  |      |
| All Major Components                                                                                                                                              | Component Name                                                                                         | Part Number         | SupplyVoltageRange | # | AbsoluteMaximumCurrent (mA)                    | TotalCurrent(mA) | Unit |
|                                                                                                                                                                   | ESP32 Microcontroller                                                                                  | ESP32-S3-WROOM-1-N4 | -0.3V to 3.6V      | 1 | 335                                            | 335              | mA   |
|                                                                                                                                                                   | OLED Screen                                                                                            | GME12864-17         | -0.3V to 4V        | 1 | 40                                             | 40               | mA   |
| B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.                                  |                                                                                                        |                     |                    |   |                                                |                  |      |
| +3.3V Power Rail                                                                                                                                                  | Component Name                                                                                         | Part Number         | SupplyVoltageRange | # | AbsoluteMaximumCurrent (mA)                    | TotalCurrent(mA) | Unit |
|                                                                                                                                                                   | ESP32 Microcontroller                                                                                  | ESP32-S3-WROOM-1-N4 | -0.3V to 3.6V      | 1 | 335                                            | 335              | mA   |
|                                                                                                                                                                   | OLED Screen                                                                                            | GME12864-17         | -0.3V to 4V        | 1 | 40                                             | 40               | mA   |
|                                                                                                                                                                   |                                                                                                        |                     |                    |   |                                                |                  |      |
|                                                                                                                                                                   |                                                                                                        |                     |                    |   | Subtotal                                       | 375              | mA   |
|                                                                                                                                                                   |                                                                                                        |                     |                    |   | Safety Margin                                  | 25%              |      |
|                                                                                                                                                                   |                                                                                                        |                     |                    |   | Total Current Required on +3.3V Rail           | 468.75           | mA   |
|                                                                                                                                                                   |                                                                                                        |                     |                    |   |                                                |                  |      |
| c1. Regulator or Source Choice                                                                                                                                    | 3.3V Switching Regulator                                                                               | LM2575D2T-3.3R4G    | 4.75V to 40V       | 1 | 1000                                           | 1000             | mA   |
|                                                                                                                                                                   |                                                                                                        |                     |                    |   | Total Remaining Current Available on 3.3V Rail | 531.25           | mA   |
|                                                                                                                                                                   |                                                                                                        |                     |                    |   |                                                |                  |      |

| +9V Power Rail                                                                                                                                                                                   | Component Name                                               | Part Number      | SupplyVoltageRange | #      | AbsoluteMaximumCurrent (mA) | TotalCurrent(mA) | Unit |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------|--------------------|--------|-----------------------------|------------------|------|
|                                                                                                                                                                                                  | 3.3V Switching Regulator                                     | LM2575D2T-3.3R4G | 4.75V to 40V       | 1      | 1000                        | 1000             | mA   |
|                                                                                                                                                                                                  |                                                              |                  |                    |        |                             |                  |      |
|                                                                                                                                                                                                  | Subtotal                                                     |                  |                    |        |                             | 1000             | mA   |
|                                                                                                                                                                                                  | Safety Margin                                                |                  |                    |        |                             | 25%              |      |
|                                                                                                                                                                                                  | Total Current Required on +9V Rail                           |                  |                    |        |                             | 1250             | mA   |
|                                                                                                                                                                                                  |                                                              |                  |                    |        |                             |                  |      |
| c2. Regulator or Source Choice                                                                                                                                                                   | DC Power Jack                                                | PJ-102AH         | 24V                | 1      | 5000                        | 5000             | mA   |
|                                                                                                                                                                                                  | Total Remaining Current Available on +9V Rail                |                  |                    |        |                             | 3750             | mA   |
|                                                                                                                                                                                                  |                                                              |                  |                    |        |                             |                  |      |
| C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above |                                                              |                  |                    |        |                             |                  |      |
|                                                                                                                                                                                                  |                                                              |                  |                    |        |                             |                  |      |
| D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need |                                                              |                  |                    |        |                             |                  |      |
| External Power Source 1                                                                                                                                                                          | Component Name                                               | Part Number      | SupplyVoltageRange | Output | AbsoluteMaximumCurrent (mA) | TotalCurrent(mA) | Unit |
| Power Source 1 Selection                                                                                                                                                                         | Plug-in Wall Supply                                          | B09ZTKTLGW       | 110VAC             | 12V    | 15000                       | 15000            | mA   |
|                                                                                                                                                                                                  |                                                              |                  |                    |        |                             |                  |      |
| Power Rails Connected to External Power Source 1                                                                                                                                                 | DC Power Jack                                                | PJ-102AH         | 24V                | 9V     | 5000                        | 5000             | mA   |
|                                                                                                                                                                                                  | 3.3V Switching Regulator                                     | LM2575-3.3WU-TR  | 4V - 40V           | 1      | 1000                        | 1000             | mA   |
|                                                                                                                                                                                                  | Total Remaining Current Available on External Power Source 1 |                  |                    |        |                             | 9000             | mA   |
|                                                                                                                                                                                                  |                                                              |                  |                    |        |                             |                  |      |